

$$C_{dG\phi_1}^{i1-i2-} \rightarrow \frac{1}{16\pi^2}$$

$$\begin{aligned} & \left(-\frac{1}{18} g_3 m_1 \tilde{\mu} g_1^2 y_d^{i1i2} \text{LF}_{2,1,1,0} [m_1, m_{\tilde{d}}^{i2}, m_{\tilde{q}}^{i1}] + \frac{1}{18} g_3 m_1 \tilde{\mu} g_1^2 y_d^{i1i2} \text{LF}_{3,1,1,-1} [m_1, m_{\tilde{d}}^{i2}, m_{\tilde{q}}^{i1}] - \right. \\ & \frac{1}{6} g_3 m_1 \tilde{\mu} g_1^2 y_d^{i1i2} \text{LF}_{2,1,1,0} [m_1, m_{\tilde{d}}^{i2}, \tilde{\mu}] + \frac{1}{6} g_3 m_1 \tilde{\mu} g_1^2 y_d^{i1i2} \text{LF}_{3,1,1,-1} [m_1, m_{\tilde{d}}^{i2}, \tilde{\mu}] - \\ & \frac{1}{12} g_3 m_1 \tilde{\mu} g_1^2 y_d^{i1i2} \text{LF}_{2,1,1,0} [m_1, m_{\tilde{q}}^{i1}, \tilde{\mu}] + \frac{1}{12} g_3 m_1 \tilde{\mu} g_1^2 y_d^{i1i2} \text{LF}_{3,1,1,-1} [m_1, m_{\tilde{q}}^{i1}, \tilde{\mu}] + \\ & \frac{1}{6} g_3 m_1 \tilde{\mu} g_1^2 y_d^{i1i2} \text{LF}_{2,2,1,-1} [m_1, \tilde{\mu}, m_{\tilde{d}}^{i2}] + \frac{1}{12} g_3 m_1 \tilde{\mu} g_1^2 y_d^{i1i2} \text{LF}_{2,2,1,-1} [m_1, \tilde{\mu}, m_{\tilde{q}}^{i1}] - \\ & \frac{3}{4} g_3 m_2 \tilde{\mu} g_2^2 y_d^{i1i2} \text{LF}_{2,1,1,0} [m_2, m_{\tilde{q}}^{i1}, \tilde{\mu}] + \frac{3}{4} g_3 m_2 \tilde{\mu} g_2^2 y_d^{i1i2} \text{LF}_{3,1,1,-1} [m_2, m_{\tilde{q}}^{i1}, \tilde{\mu}] + \\ & \frac{3}{4} g_3 m_2 \tilde{\mu} g_2^2 y_d^{i1i2} \text{LF}_{2,2,1,-1} [m_2, \tilde{\mu}, m_{\tilde{q}}^{i1}] - \frac{1}{6} m_3 \tilde{\mu} g_3^3 y_d^{i1i2} \text{LF}_{2,1,1,0} [m_3, m_{\tilde{d}}^{i2}, m_{\tilde{q}}^{i1}] - \\ & \frac{4}{3} m_3 \tilde{\mu} g_3^3 y_d^{i1i2} \text{LF}_{3,1,1,-1} [m_3, m_{\tilde{d}}^{i2}, m_{\tilde{q}}^{i1}] - \frac{1}{6} g_3 m_1 \tilde{\mu} g_1^2 y_d^{i1i2} \text{LF}_{2,1,1,0} [\tilde{\mu}, m_1, m_{\tilde{d}}^{i2}] + \\ & \frac{1}{6} g_3 m_1 \tilde{\mu} g_1^2 y_d^{i1i2} \text{LF}_{3,1,1,-1} [\tilde{\mu}, m_1, m_{\tilde{d}}^{i2}] - \frac{1}{12} g_3 m_1 \tilde{\mu} g_1^2 y_d^{i1i2} \text{LF}_{2,1,1,0} [\tilde{\mu}, m_1, m_{\tilde{q}}^{i1}] + \\ & \frac{1}{12} g_3 m_1 \tilde{\mu} g_1^2 y_d^{i1i2} \text{LF}_{3,1,1,-1} [\tilde{\mu}, m_1, m_{\tilde{q}}^{i1}] - \frac{3}{4} g_3 m_2 \tilde{\mu} g_2^2 y_d^{i1i2} \text{LF}_{2,1,1,0} [\tilde{\mu}, m_2, m_{\tilde{q}}^{i1}] + \\ & \frac{3}{4} g_3 m_2 \tilde{\mu} g_2^2 y_d^{i1i2} \text{LF}_{3,1,1,-1} [\tilde{\mu}, m_2, m_{\tilde{q}}^{i1}] + \frac{1}{2} g_3 \tilde{\mu} y_d^{pi2} y_u^{i1r} \overline{a_u^{pr}} \text{LF}_{2,1,1,0} [\tilde{\mu}, m_{\tilde{q}}^p, m_{\tilde{u}}^r] - \\ & \left. \frac{1}{2} g_3 \tilde{\mu} y_d^{pi2} y_u^{i1r} \overline{a_u^{pr}} \text{LF}_{3,1,1,-1} [\tilde{\mu}, m_{\tilde{q}}^p, m_{\tilde{u}}^r] \right) \end{aligned}$$